

Light-cone-quantized QCD in 1+1 dimensions, revisited

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I. INTRODUCTION

In 1990, Hornbostel, Brodsky and Pauli diagonalized the light-cone Hamiltonian of $SU(N)$ gauge theory in one space and one time dimension in a discrete momentum basis, and computed the hadronic spectrum and wavefunctions across couplings, numbers of colours, and baryon number [1]. The calculation ran at resolutions $2K = 16\text{--}24$, in minutes on the mainframes of the day, with convergence controlled by a single extrapolation formula. It became a standard reference for discretized light-cone quantization (DLCQ), and its figures are still the picture most physicists carry of what QCD in two dimensions looks like at finite N : which states exist, how their wavefunctions deform with coupling, how quickly large- N becomes a good approximation.

The reasons to study the model have, if anything, strengthened since. Tractability remains: the theory confines, has a meaningful chiral limit, and carries one dimensionless parameter, while being solvable to high precision. The role of test bed has changed hands — where the 1990 paper looked ahead to DLCQ in four dimensions, today the same model is the standard benchmark for Hamiltonian lattice methods, tensor networks, and quantum simulation, which makes controlled finite- N spectra directly useful to communities that did not exist then. And the model has open questions of its own: the chiral-limit behaviour at finite N is measured differently by light-front and equal-time methods (Sec. X), and the existence of compact multiquark states near threshold is an active large- N prediction awaiting a finite- N test [2, 3].

This paper re-does the 1990 calculation with thirty-six years of hardware and — more importantly — of error analysis. We reproduce every figure and the table of that paper, at the original parameters and again at resolutions up to $2K \simeq 100$, from two independent implementations of the same Hamiltonian: the preserved Fortran of the original program, and a validated open-source port. Where the original controlled its errors with one extrapolation series and a parenthesis convention, we carry a four-component error budget with held-out validation (Sec. VII); where its weak-coupling entries were limited by the momentum grid in a way no resolution can repair, we identify the mechanism, quantify it without free parameters, and reach the chiral region instead with van de Sande’s improved Hamiltonian [4] (Sec. VIII). Two sections have no 1990 counterpart: a search for states dom-

inated by four- and six-parton Fock sectors — tetraquark and hexaquark candidates — (Sec. IX), and a comparison against the modern record: integrability results at large N , tensor-network and quantum-hardware spectra at $N = 2, 3$, and precision Schwinger-model benchmarks (Sec. X).

We have kept the structure of Ref. [1] deliberately, section for section where possible, so that each figure here can be laid beside its 1990 counterpart. Sections II and III fix notation and recall the Hamiltonian; Sec. IV describes what changed computationally; Secs. V–VI reproduce and extend the 1990 results; the remainder is new.

II. LIGHT-CONE QUANTIZATION

We follow the conventions of Ref. [1] throughout. Light-cone coordinates are $x^\pm = x^0 \pm x^1$, with x^+ playing the role of time; the conjugate momenta are $p^\mp$, and the mass-shell condition $p^+p^- = m^2$ makes p^- the light-cone energy of a quantum carrying longitudinal momentum $p^+ > 0$. Positivity of p^+ is the structural advantage of the scheme: no combination of quanta with strictly positive longitudinal momenta can add up to the vacuum’s $p^+ = 0$, so the Fock vacuum is an exact eigenstate of the full Hamiltonian and every basis state is built from quanta that belong to the hadron, not to vacuum fluctuations. The familiar equal-time picture, in which even a two-particle state mixes with arbitrarily complicated vacuum structure, is replaced by one in which the partition of a single conserved quantity organizes the entire basis.

Discretization proceeds by placing the theory in a box of length $2L$ in x^- with antiperiodic boundary conditions for the fermions, so that momenta come in half-odd-integer units $k_n^+ = 2\pi n/L$, $n = \frac{1}{2}, \frac{3}{2}, \dots$. The dimensionless resolution

$$K = \frac{L}{2\pi} P^+ \quad (1)$$

is a positive integer (in code units we track $2K$, which is odd or even as the parton number is), and a basis state is nothing but a colour-singlet partition of K among quanta. At $K = 3$, for example, the available momentum fractions are the partitions (3), (2, 1) and (1, 1, 1) — to be contrasted with the unbounded sets of positive and negative momenta an equal-time discretization admits at the same volume. Because the smallest available momentum fraction is $x = k/K \sim 1/K$, the resolution doubles as an infrared regulator of the $x \rightarrow 0$ endpoint, and the continuum limit $L \rightarrow \infty$ at fixed P^+ is the single limit $K \rightarrow \infty$. How observables approach that limit — and

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when the $1/K$ endpoint cutoff is the dominant error — is the subject of Secs. VII and VIII.

Everything else — the mode expansions, the elimination of the constrained degrees of freedom, and the construction of P^- — is unchanged from Ref. [1], and we do not repeat it.

III. SU(N) IN 1+1 DIMENSIONS AND THE IMPROVED HAMILTONIAN

The theory is SU(N) gauge theory coupled to a single fundamental fermion of mass m ,

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} + \bar{\psi}(i\cancel{D} - m)\psi, \quad (2)$$

in one space and one time dimension, where the coupling g carries dimensions of mass and m/g is the only physical parameter. In the gauge $A^+ = 0$ the gluon field carries no dynamical degrees of freedom: A^- is fixed by a constraint equation, and integrating it out leaves quarks interacting through an instantaneous linear potential in x^- — confinement is kinematic in two dimensions, in QED₁₊₁ as much as here. Half of the fermion components are likewise constrained, so the physical content is a single spinless quark field per colour.

The light-cone Hamiltonian then takes the form

$$P^- = m^2 \sum_{n,c} \frac{a_{nc}^\dagger a_{nc}}{k_n^+} + g^2 H_I, \quad (3)$$

where H_I contains the normal-ordered four-quark vertices generated by the instantaneous potential, with the colour structure $\delta_{c_1 c_3} \delta_{c_2 c_4} - \frac{1}{N} \delta_{c_1 c_2} \delta_{c_3 c_4}$ and a $1/(\Delta k^+)^2$ momentum kernel, together with a one-body “self-induced inertia” — the normal-ordering remnant of the instantaneous exchange, diagonal in the Fock basis. Explicit matrix elements are given in Ref. [1]; our implementations of them, in both the preserved 1988-era Fortran and an independently validated Python port, are public (Appendix D).

III.1. The endpoint problem and the improved Hamiltonian

Near the endpoints $x \rightarrow 0, 1$ the continuum wavefunction of a bound state behaves as $\phi \sim x^a$, with the exponent set by the transcendental condition [1, 5]

$$1 - \pi a \cot(\pi a) = \frac{\pi m^2}{g^2 C_F}, \quad C_F = \frac{N^2 - 1}{2N}. \quad (4)$$

As $m/g \rightarrow 0$ the exponent $a \rightarrow 0$ and the endpoint region $x \lesssim e^{-1/a}$ shrinks below any fixed grid: a uniform-in- k^+ discretization whose smallest momentum fraction is $1/K$ then replaces the endpoint integral $\int dx x^{2a-1} \sim 1/2a$ by a logarithm $\ln K$, and no reachable resolution recovers it (Sec. VII). This is a property of the grid, not of light-cone

quantization: a continuum ‘t Hooft solver on a uniform mesh, sharing no code with the discretized theory, fails in exactly the same way.

Van de Sande’s improved Hamiltonian [4] treats the disease at its source. The self-induced inertia and the Coulomb exchange suffer separate, cancelling endpoint singularities; the improvement replaces the free-field self-energy by one computed against the interacting endpoint behaviour of Eq. (4), so that the residual kernel vanishes when either momentum approaches an endpoint. Two properties make this essentially free within an existing DLCQ code. First, the modification is exactly diagonal: it amounts to swapping the self-energy table $\sigma_{\text{std}} \rightarrow \sigma_{\text{imp}}$, leaving every interaction kernel — and the standard path, bit for bit — untouched. Second, the colour weight it requires is $\langle -T_a \cdot T_c \rangle$ at zero momentum transfer, which the colour-contraction machinery already computes; the identity $\sum_{c \neq a} \langle -T_a \cdot T_c \rangle = C_F$ holds exactly in our implementation at every N and parton number we use, and is enforced as a gate rather than assumed.

One rule attaches to everything that follows and is worth stating once, prominently. The two Hamiltonians converge to the continuum along different paths: the improved spectrum is extrapolated in a plain $1/K$ series [4], while the standard spectrum requires the fractional-power ladder of Ref. [1], Eq. (27) there. The two bases are never mixed, and no fit in this paper crosses them; the error budget in Sec. VII quantifies what happens when one does.

IV. THE NUMERICAL METHOD

V. WAVE FUNCTIONS AND STRUCTURE FUNCTIONS

VI. MASSLESS MESONS AND BARYONS IN THE $m/g \rightarrow 0$ LIMIT

At $m = 0$ the spectrum contains towers of exactly massless hadrons, and the construction — due to Ref. [1], Sec. VI — survives unchanged, so we record only what this paper uses it for. The colour currents V_k^a built from the quark bilinears obey a Kac–Moody algebra, and the interacting part of P^- can be rewritten as a quadratic form in them; states created purely by currents (and, for baryon number, by a composite baryon operator) are annihilated by it, giving massless mesons, baryons, and multi-hadron states at *every* resolution K , not merely in the limit. The valence distribution of the massless baryon is known in closed form,

$$q(x) = N(N-1)(1-x)^{N-2}, \quad (5)$$

which at $N = 2$ collapses to the flat meson distribution. For U(N) the extra U(1) current promotes these states to free massive bosons at $M^2 = g^2/2\pi$ — the Schwinger mechanism — which is the sharpest statement of how SU(N) and U(N) differ in two dimensions.

Three later results lean on this section. The identically-zero last row of Table ?? is exact and K -independent, and our solvers reproduce it as a structural check at every resolution. The chiral-region analysis of Sec. VIII uses masslessness as the fixed endpoint that any $M^2(m/g)$ fit must respect. And the exotics search of Sec. IX inherits a boundary condition: since multi-hadron states are exactly massless at $m = 0$, every binding energy we measure must vanish in the chiral limit, and a candidate whose Δ fails to do so is diagnosing its own systematics, not new physics.

VII. CONVERGENCE AND THE ERROR BUDGET

VIII. THE CHIRAL REGION

IX. TETRAQUARK AND HEXAQUARK CANDIDATES

X. COMPARISON WITH MODERN METHODS

The 1990 paper compared against the two external results then available: 't Hooft's large- N solution and Hamer's finite-lattice SU(2) spectrum [6]. The modern record is far richer, and this section is organized by what each method can check.

X.1. Large N and integrability

The 't Hooft equation is now solved to essentially arbitrary precision: its equivalence to a TQ-Baxter system [7, 8] turns the large- N meson spectrum into a benchmark exact for our purposes. Our finite- N masses, rescaled to 't Hooft's normalization (Appendix C), approach these values with the expected $1/N^2$ systematics, updating Fig. 8(b) of Ref. [1] with error bars on both axes' inputs. The same rescaled comparison exposes what large N misses at strong coupling — light baryons, and the U(1) distinction of Sec. VI — exactly as the original argued, now with the residuals quantified.

X.2. Lattice

Hamer's SU(2) finite-lattice masses remain the only equal-time spectrum across the full coupling range at $N = 2$. We compare against his published Table 1, whose stated uncertainties at weak coupling are of order the values themselves; a digitization of his figure would suggest a precision his own table disclaims, and we deliberately do not use one. Within those bars, agreement is complete, including the $N = 2$ identity $M_{\text{bar}} = M_{\text{mes}}$ that his data was the first to exhibit. Modern overlap-fermion lattice work confirms the expected infrared behaviour at $(N_c, N_f) = (2, 1)$ and $(3, 1)$ [9], though without spectrum tables to compare digit-for-digit.

X.3. Tensor networks and quantum hardware

Hamiltonian-lattice calculations with matrix product states now reach SU(2) [10, 11] and SU(3) [12, 13] in one dimension, and variational quantum hardware has produced SU(2) meson and baryon masses [14] with SU(3) preparations following [15]. These are staggered-fermion, finite-lattice-spacing computations, so the honest comparison is at matched continuum extrapolations rather than raw numbers; where the published works quote continuum values, we tabulate them beside ours. Precision benchmarks in the Schwinger model [16, 17] calibrate what tensor-network methods achieve at U(1), and our equal-time exact-diagonalization cross-checks (Appendix D) reproduce them in our own conventions. No public code exists for any of the SU(N) tensor-network calculations above; the open pipeline released with this paper is, to our knowledge, the first reproducible finite- N spectrum code in either quantization.

X.4. The chiral-exponent tension

One discrepancy in this landscape is structural, and we state it as the open question it is. Every light-front determination of the finite- N chiral exponent — this work, lightcone conformal truncation [18], and $1/N$ -corrected 't Hooft perturbation theory [19] — finds $M^2 \propto (m/g)^\alpha$ with $\alpha \rightarrow 1$, while the one equal-time measurement, an MPS study of SU(2) over $m/g \in [0.1, 0.4]$ [11], reports $\nu = 0.700(29)(11)$ for $M \propto (m/g)^\nu$, i.e. $\alpha \simeq 1.4$: consistent with the bosonization prediction and several standard deviations from the light-front value. The methods share no assumption more specific than quantization itself — but they differ in precisely the one all light-front schemes share, the trivial vacuum, which is what an anomalous chiral exponent is about. The equal-time side of the ledger is not yet decisive either: the measured window spans a factor of four in m/g , not a decade, and no SU(N) analogue of the additive staggered mass shift [17] — the same order in the lattice spacing as the mass being fitted — has been published or applied. Resolving the tension requires an equal-time computation in the same chiral window with that renormalization under control; it is beyond the published record, and beyond this paper.

XI. PROSPECTS

Four extensions are within reach of the machinery described here, in rough order of readiness.

QCD coupled to QED with physical charges. The colour-contraction tables already compute both Fierz structures, and the U(1) exchange reuses the $-1/N$ index structure with different weights, so $H = H_{\text{QCD}} + (e^2/g^2)H_{\text{QED}}$ is available as a reweighting of matrix elements already built, with one build serving every charge ratio. The validation ladder is clean — bit-identity at $e = 0$, the exact

Schwinger mass at the $U(1)$ point, isospin restoration — but Gauss’ law on a circle admits only neutral states without a zero-mode subtraction: the neutron and π^0 come free, the proton does not.

The $U(N)$ switch. The same reweighting at $e^2/g^2 \rightarrow$ pure $U(1)$ limit exposes the $SU(N)/U(N)$ split of Sec. VI numerically; the thesis [20] plots the $SU(2)$ – $U(2)$ comparison that the 1990 article never showed, and reproducing it would close the last unreproduced published figure of this program.

Zero modes. Our antiperiodic quantization removes the $k^+ = 0$ mode by fiat. In two-dimensional models where zero modes have been included dynamically, low-lying masses shift by as much as $\sim 20\%$ [21], so their absence is a genuine systematic of unknown size for this paper’s absolute masses — though not the cause of the weak-coupling failure, which a continuum solver with no zero-mode truncation reproduces exactly (Sec. VII). Collective-variable treatments that tensor a zero-mode sector onto the Fock basis exist and are compatible with our solver architecture.

An equal-time computation in the same chiral window. The sharpest open discrepancy this paper documents — light-front methods finding $M^2 \propto (m/g)^\alpha$ with $\alpha \rightarrow 1$ where the one equal-time measurement suggests $\alpha \approx 1.4$ (Sec. X) — will not be settled by more K . It needs an equal-time calculation with a controlled additive mass renormalization at finite N , over a decade in m/g rather than a factor of four. The missing theoretical ingredient is the $SU(N)$ analogue of the staggered-fermion mass shift known for the Schwinger model [17]; deriving it is, in our view, the single most valuable piece of new work available in this corner of field theory, and the natural subject of a companion study.

Appendix A: Error-budget definitions

Appendix B: Approaches measured not to work

Every extrapolation strategy below was implemented and measured against the standard-Hamiltonian weak-coupling problem of Sec. VII before being rejected. We record them because each is the obvious next idea, and each fails in a way that is invisible from inside the fit — the same property that makes the underlying problem dangerous. Numbers refer to the Table ?? channels at $m/g \leq 0.2$ unless stated.

More resolution.: The remainder falls only as K^{-a} with $a \rightarrow 0$; halving the uncertainty bracket at $m/g = 0.1$ requires increasing K by a factor of order $10^{4.9}$, and at the smallest couplings we probe, by factors that exceed any conceivable computation. Resolution buys nothing below the validity boundary; that is the point.

A confluent fit basis.: Replacing the near-degenerate $\{K^{-1}, K^{-(1+a)}\}$ pair by a confluent basis spans the

same space: the extrapolated mass is identical to one part in 10^{14} .

Committing to a single series order.: Fixing the number of terms (say four) hides, not removes, the order dependence: the next admissible term still moves the extrapolated mass by several percent. Two defensible selection criteria — held-out resolution and consistency with the known chiral behaviour — prefer opposite ends of the admissible range, which is what an undetermined model order looks like; the budget therefore spans the whole ladder rather than choosing.

A validity floor on the fit windows.: Raising the smallest resolution admitted to the window set shrinks the quoted spread severalfold while walking the central value monotonically by many times the shrunken bar — variance traded for bias, silently.

Fitting the endpoint exponent freely.: $M_0 - C K^{-p}$ is unidentifiable when the true p is small: K^{-p} is nearly constant, M_0 and C degenerate, and the fit lands an order of magnitude from the true value with a residual of 10^{-7} — a perfect fit to the wrong answer.

Summing the tail analytically.: A Hurwitz- ζ resummation of the increment series validates at strong coupling and diverges exactly where it is needed.

Adding an explicit K^{-a} column.: With a fixed from Eq. (4), the extrapolated masses land several times above the published values while the fitted chiral exponent stays pinned to the grid artifact; inside a joint $(m/g, K)$ fit the same column produces variant spreads of hundreds of percent and negative masses.

The common failure mode deserves the emphasis: every rejected strategy *improved* some internal fit diagnostic while degrading the answer. Residuals, held-out predictions, and spread statistics all reward flexibility inside the reachable window, and the reachable window does not contain the physics (Sec. VII). Where we quote weak-coupling numbers from the standard Hamiltonian, the error bar carries the endpoint systematic in full; where we need the chiral region, we change the Hamiltonian, not the fit.

Appendix C: Units and conventions

Masses are quoted as M/g unless stated otherwise; m/g is the quark mass in units of the coupling. The natural combination in this paper’s normalization is $g^2 N/2\pi$, whereas ’t Hooft’s classic solution uses $g^2 N/\pi$; the large- N comparison in Sec. X therefore plots $(2\pi/N)^{1/2} M/g$ against $(2\pi/N)^{1/2} m/g$, exactly as Fig. 8(b) of Ref. [1] did. Spectra shown as functions of the coupling use the compactified variable $\lambda = [1 + \pi m^2/g^2]^{-1/2}$, which maps

the free limit to $\lambda \rightarrow 0$ and the chiral limit to $\lambda = 1$, and the dimensionless mass $M^2/(m^2 + g^2/\pi)$, both directly comparable to the 1990 figures. In code units the resolution is $2K$ (integer, matching the parity of the parton number) and single-parton momenta are odd integers $k = 1, 3, \dots$, so a structure function lives on $x = k/2K$ with measure $dx = 1/K$.

Two conventions of Ref. [1] are traps for anyone comparing against it, and we state them explicitly since both bit us. First, *the numbers printed in its Table I are $M^2/(m^2 + g^2/\pi)$, although the column headers read M/g* . The two agree at no coupling; the identification was established by recomputing both quantities at the 1990 parameters and matching digit patterns, and our transcription of the table carries the correction with full provenance. Every comparison in this paper converts to genuine M/g , and the cross-check that our conversion is right — recomputing the comparison in the table’s native M^2 units — is part of the build (Appendix D). Second, the parenthesized digits in that table are *not* error bars: they record the magnitude of the last term retained in the extrapolation series, a convergence diagnostic the authors themselves trusted only for $m/g \gtrsim 0.2$. Where this paper quotes an uncertainty, it is the full budget of Sec. VII;

where it quotes the 1990 numbers, the parenthesis keeps its original meaning.

Captions in this paper state N , B , $2K$ (or the fit window), m/g , the Fock-space truncation, and the Hamiltonian, for every panel; where the original figure left a parameter unstated, the value we recovered and the evidence for it are cited inline.

Appendix D: Data and code availability

Both solvers — the preserved 1988-era Fortran, unmodified, and the Python port — are public, together with every analysis script, the digitized markers of all 1990 figure panels with their axis-calibration provenance, and the per-figure data files, each carrying in its header the exact command that regenerates it. Every number quoted in this paper’s prose is a macro generated from those data files by a single extractor, with the repository revision and input hashes recorded alongside; every figure is built by a single driver that reads cached solver output and refuses to solve, so that a missing input surfaces as an explicit run command rather than a silent recomputation. The equal-time exact-diagonalization used for the Schwinger-model cross-checks of Sec. X is included.

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